



AnalystLab Africa - HealthConnect Experience Lab

Data Science → Data Analytics: Week 7 Validation Response

In response to your request to cross-validate Power BI outputs against independently calculated Data Science results, the confirmed figures below come from the same HealthConnect appointment dataset.

1. Core KPIs

KPI	Value (n = 5,000)
Total appointments	5,000
Unique patients	1,696
Attendance rate	46.3%
Overall no-show rate	48.5%
Cancellation rate	5.3%

These match your dashboard exactly.

2. Key Analytical Findings

Denominator = Attended + No-Show only (n = 4,737); Cancelled appointments excluded, consistent with the Week 6 modelling approach.

Finding	Data Science Result	Your Original Figure
No-show rate - 0 previous no-shows	46.3%	43.5%
No-show rate - 1 previous no-show	55.9%	53.5%
No-show rate - 2 previous no-shows	62.1%	59.4%
No-show rate - 3+ previous no-shows	70.3%	68.8%
No-show rate - 31-60 day lead time	63.9%	60.5%
No-show rate - reminder sent	49.9%	47.4%
No-show rate - reminder not sent	54.6%	51.4%
No-show rate - distance ≥ 20 km	60.2%	~58%

All figures are close to your originals - small differences likely reflect minor denominator or rounding differences, but the underlying patterns match closely. Good cross-validation between our two independent approaches.

3. Week 7 Segment Validation: Does the 31-60 Day Pattern Hold Across Appointment Types?

Yes - the elevated no-show risk in the 31-60 day lead-time band is consistent across all four appointment types, not concentrated in any one type:

Appointment Type	No-Show Rate (31-60 day band)
Follow-up	68.8% (highest)
Specialist Consultation	63.0%
Diagnostic Test	62.5%
General Consultation	61.4%

This is a broad-based pattern rather than one driven by a single appointment type.

4. Data Science Week 7 Findings Relevant to These Variables

- Cross-track validation: the Data Science model's top-risk predictions align 98.3% with your independently-defined high-risk segments (3+ previous no-shows, 31-60 day lead time, distance ≥ 20 km) strong mutual validation of both approaches.
- Important flag: a gender fairness gap was discovered in the model (ROC-AUC 0.725 for male patients vs 0.639 for female) that none of the four variables above directly explain. An attempted fix (gender-interaction features) did not close the gap.
- Request: if you have bandwidth, it would help to check whether any of the four variables above (previous no-shows, lead time, reminders, distance) show a different pattern by gender in your dashboard this could help explain what the model is currently missing.

Happy to provide any further detail needed for your Week 7 documentation.